

private sector bodies but this still leaves room for bias and corruption because employees might not report all irregularities.

4: UNDERSTAND DECISION TABLES.

4.1 Define Decision Table

A Decision Table: - is a tabular representation of inputs versus rules/cases/test conditions. It is a very effective tool used for both complex software testing and requirements management. A decision table helps to check all possible combinations of conditions for testing and testers can also identify missed conditions easily. The conditions are indicated as True(T) and False(F) values.

What is Decision Table Testing:- Decision table testing is a software testing technique used to test system behavior for different input combinations. This is a systematic approach where the different input combinations and their corresponding system behavior (Output) are captured in a tabular form. That is why it is also called as a Cause-Effect table where Cause and effects are captured for better test coverage.

Why Decision Table Testing is Important?

Decision Table Testing is important because it helps to test different combinations of conditions and provides better test coverage for complex business logic. When testing the behavior of a large set of inputs where system behavior differs with each set of inputs, decision table testing provides good coverage and the representation is simple so it is easy to interpret and use.

Advantages of Decision Table Testing

- ❖ When the system behavior is different for different inputs and not the same for a range of inputs, both equivalent partitioning and boundary value analysis won't help, but a decision table can be used.
- ❖ The representation is simple so that it can be easily interpreted and is used for development and business as well.
- ❖ This table will help to make effective combinations and can ensure better coverage for testing.
- ❖ Any complex business conditions can be easily turned into decision tables.
- ❖ In a case we are going for 100% coverage typically when the input combinations are low, this technique can ensure the coverage.

Disadvantages of Decision Table Testing

- ❖ The main disadvantage is that when the number of inputs increases the table will become more complex.

4.2 Explain Result table.

The data that is retrieved by an SQL statement is always in the form of a table, which is called a RESULT TABLE. Like the tables from which you retrieve the data, a result table has rows and columns. A program fetches this data one row at a time.

What does table mean in research?

Tables are used to organize data that is too detailed or complicated to be described adequately in the text, allowing the reader to quickly see the results. They can be used to highlight trends or patterns in the data and to make a manuscript more readable by removing numeric data from the text.

4.3 Explain Quadrant table.

What is a Quadrant:- The word "quad" means four in Latin. This meaning ties into what a quadrant is. On a graph, a quadrant is one of the four sections created by the x-axis and y-axis crossing at a 90-degree angle. This intersection of the axes forms four equal quadrants. Each

quadrant is surrounded by the x-axis and the y-axis producing different sets of ordered pairs. The ordered pairs provide not only a location in reference to (0, 0), their coordinates are also indicators to which quadrant the pair is located.

A table: - is a collection of related data held in a table format within a database. It consists of columns and rows.

What are the Quadrants of a Graph? The quadrants of a graph consist of four sections based on both positive and negative coordinates for x and y. The combination of the x-signs. So what are these four sections, how are they labeled, and what are their properties?

5: UNDERSTAND PROJECT PLANNING AND SCHEDULING.

5.1 What is Planning & Scheduling?

What is planning: - Planning is one of the first stages of project development, in which companies and businesses determine how to achieve a specific goal or set of goals. During the planning phase, project managers create an action plan that helps them determine what steps are necessary to complete before they can achieve the project's goal. Companies use this time to develop a structure by which they can enact an initiative, such as a product update, campaign, policy change or partnership.

What is scheduling: - Scheduling is the act of determining who might accomplish certain steps of a project's action plan and when they might do so. They create a time-based sequence to complete the project. During the scheduling phase, companies examine their resources and determine which ones are necessary to complete the project's goal. Project leaders may reference the budget determined in the planning stage and allocate resources, including equipment, team members and time.

Differences between Planning & Scheduling.

S/N	CRITERIA	PLANNING	SCHEDULING
1	Goal	Define Scope, Component, Activities, Resources, Policies, Procedures & Potential Risk of the Project.	Define Task, Owners, Timeline orders of events, Progress, Benchmark & Risk Management Measures.
2	Timing	Occurs before project begins.	Occurs soon after project begins.
3	Reliance	Strategy.	Plans.
4	Term	Long-term initiatives.	Short-term initiatives.

5.1 Explain the techniques for project planning and scheduling.

- 1. Plan schedule management:** - The groundwork for a good project schedule is to establish the procedures, company policies, and documentation guidelines that will govern your project. The plan for schedule management outlines resources available for the project and the contingencies that may arise.
- 2. Define the project activities:** - This can be as simple as creating a list of tasks that must be completed in order to deliver your project. In the case of complex projects, it may be helpful to organize these tasks in the form of a chart visualizing project tasks and their sub-tasks and to stay organized at work.
- 3. Determine dependencies:** - Once you have all the project activities listed, think through each one carefully to identify which tasks rely on others to be completed. If you're building a house, for example, you can't put the roof on until the frame is completed. It's important to correctly define all your project dependencies so you can schedule accurately and avoid project delays.

4. **Sequence activities:** - After you've established dependencies among your activities, you can sequence them. At this point, you aren't assigning any *time* to your activities in terms of work hours or due dates. Instead, you're focusing on the *order* in which all project activities should be done so that the most efficient flow is created.

5. **Estimate resources:** - Each activity in your project will require resources in the form of personnel, subcontractor costs, tools (physical and/or digital tools like software programs), and workspace. Make sure to consider other resources that are specific to your industry or project. Estimate the resources needed for each project activity.

6. **Estimate durations:** - This step is pretty obvious but very important. *How long will each project activity take?* Underestimating will, of course, put you behind schedule and ultimately frustrate your customer.

(Overestimating could leave team members or other resources sitting idle as they wait for antecedent tasks to be completed. The best way to estimate duration is to use data from similar previous jobs.

If you don't have any data to work from and there's no industry standard to which you can refer, an estimate based on the average of the best, worst, and most likely scenarios.

7. **Develop the project schedule:** - At this point, you should have all the information you need to develop your project schedule. Taking into consideration the duration and resource requirements of each activity, as well as their dependencies and proper sequence, you can assign start dates and due dates for each activity.

8. **Monitor and control:** - Unlike the rest of the project scheduling steps, Step 8 is ongoing. As a project manager, you'll be monitoring and controlling your project schedule for the duration of the project. This step involves running project reports and assessing the progress of a project against the schedule, managing performance, and communicating with the team.

5.2 Explain CPM and PERT. Solve case studies.

PERT: - is an acronym for Program (Project) Evaluation and Review Technique, in which planning, scheduling, organizing, coordinating and controlling uncertain activities take place. The technique studies and represents the tasks undertaken to complete a project, to identify the least time for completing a task and the minimum time required to complete the whole project. It was developed in the late 1950s. It is aimed to reduce the time and cost of the project.

PERT uses time as a variable which represents the planned resource application along with performance specification. In this technique, first of all, the project is divided into activities and events. After that proper sequence is ascertained, and a network is constructed. After that time needed in each activity is calculated and the critical path (longest path connecting all the events) is determined.

CPM: - Developed in the late 1950s, Critical Path Method or CPM is an algorithm used for planning, scheduling, coordination and control of activities in a project. Here, it is assumed that the activity duration is fixed and certain. CPM is used to compute the earliest and latest possible start time for each activity.

The process differentiates the critical and non-critical activities to reduce the time and avoid the queue generation in the process. The reason for the identification of critical activities is that, if any activity is delayed, it will cause the whole process to suffer. That is why it is named as Critical Path Method.

In this method, first of all, a list is prepared consisting of all the activities needed to complete a project, followed by the computation of time required to complete each activity. After that, the dependency between the activities is determined. Here, 'path' is defined as a sequence of activities in a network. The critical path is the path with the highest length.

Key Differences Between PERT and CPM

- ❖ PERT is a project management technique, whereby planning, scheduling, organizing, coordinating and controlling uncertain activities are done. CPM is a statistical technique of project management in which planning, scheduling, organizing, coordination and control of well-defined activities take place.
- ❖ PERT is a technique of planning and control of time. Unlike CPM, which is a method to control costs and time?
- ❖ PERT is evolved as a research and development project. CPM evolved as a construction project.
- ❖ PERT is set according to events while CPM is aligned towards activities.
- ❖ A deterministic model is used in CPM. Conversely, PERT uses a probabilistic model.
- ❖ There are three times estimates in PERT, i.e. optimistic time (to), most likely time (tm), pessimistic time (tp). On the other hand, there is only one estimate in CPM.
- ❖ PERT technique is best suited for a high precision time estimate, whereas CPM is appropriate for a reasonable time estimate.
- ❖ PERT deals with unpredictable activities, but CPM deals with predictable activities.
- ❖ PERT is used where the nature of the job is non-repetitive. In contrast to, CPM involves the job of repetitive nature.
- ❖ There is a demarcation between critical and non-critical activities in CPM, which is not in the case of PERT.
- ❖ PERT is best for research and development projects, but CPM is for non-research projects like construction projects.
- ❖ Crashing is a compression technique applied to CPM, to shorten the project durations along with the least additional cost. The crashing concept is not applicable to PERT.

Example 1: A small project consisting of eight activities has the following characteristics:

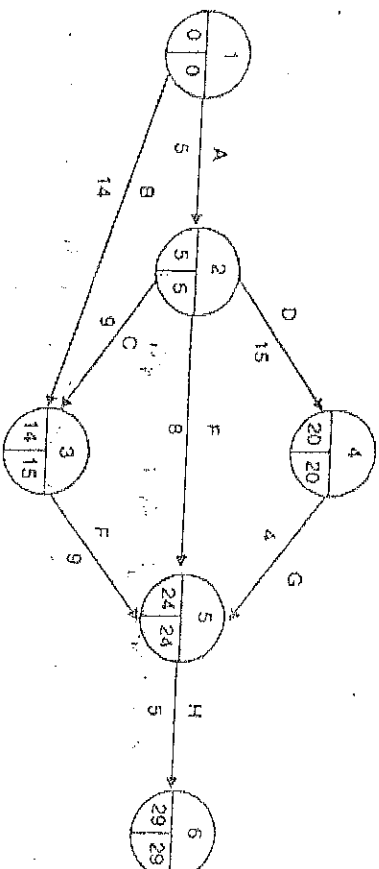
Activity	Preceding activity	Time – Estimates (in weeks)		
		Most optimistic time (to)	Most likely time (tm)	Most Pessimistic time (tp)
A	None	2	4	12
B	None	10	12	26
C	A	8	9	10
D	A	10	15	20
E	A	7	7.5	11
F	BC	9	9	9
G	D	3	3.5	7
H	E, F, G	5	5	5

- (i) Draw the PERT network for the project.
- (ii) Prepare the activity schedule for the project.
- (iii) Determine the critical path.
- (iv) If a 30- week deadline is imposed, what is the probability that the project will be finished within the time limit?

If the project manager wants to 99% sure that the project is completed on the schedule date, how many weeks before that date should he start the project work?

Solution: The network diagram for the given data is shown in fig. below. The earliest time and variance of each activity is computed by using the formula.

$$t_c = \frac{a + 4m + b}{6} \text{ and } \sigma^2 = \left(\frac{b - a}{6} \right)^2$$



PERT Network diagram

(ii) Calculation activity duration and scheduling times.

Activity	Time estimates			t_c	(σ^2)	Earliest time		Latest time	
	a	m	b			Start	Finish	Start	Finish
A	2	4	12	5	25/9	0	5	0	5
B	10	12	26	14	64/9	0	14	1	15
C	8	9	10	9	1/9	5	14	6	15
D	10	15	20	15	25/9	5	20	5	20
E	7	7.5	11	8	4/9	5	13	16	24
F	9	9	9	9	0	14	23	15	24
G	3	3.5	7	4	4/9	20	24	20	24
H	5	5	5	5	0	24	29	24	29

(iii) The critical path of the project is 1-2-4-5-6, critical activities being A, D, G and H.

The expected project length is the sum of duration of each critical activity. Expected project length $5 + 15 + 4 + 5 = 29$ weeks.

Variance project length is obtained by summing variance of each critical activity.

Variance of the project $25/9 + 25/9 + 4/9 + 0 = 6$

(iv) The required probability can be determined by finding the area under the normal curve to the left of $X = 30$

Now, the probability of completing the project within the 30 week deadline is

$$\begin{aligned}
 P(X \leq 30) &= 0.5 + P(\mu < x < 30) \\
 &= 0.5 + P(0 \leq Z \leq 0.41) \\
 &= 0.5 + 0.1591 \\
 &= 0.6591
 \end{aligned}$$

Where

$$Z = \frac{\text{Due date} - \text{Expected date}}{\sigma}$$

$$Z = \frac{30 - 29}{\sqrt{6}} = 0.41$$

(v) If the project start T weeks before the due date, the X will represent the ordinate under normal curve to the left of which 99% of area lies.

The area between n and X -being 99.50 or 49% and Z value corresponding to this is 2.03 λ (from table)

$$2.33 = \frac{T - 29}{\sqrt{6}}$$

$$T = 29 + 2.33 \sqrt{6} \\ = 34.7 \text{ weeks}$$